
Introduction to Algorithms

PHIL 3050: Information and Uncertainty

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An algorithm can be [defined](#) as “a set of instructions that is designed to accomplish a task.” A recipe for baking cookies, a program to find a particular value in a list of values, and a program to write a poem about a cat are all examples of algorithms. The code covered in this module is primarily relevant to searching algorithms.

Searching Algorithms

[Searching algorithms](#) are “used to find an element or a value within a collection of data.”

For example, the below code uses a `for` loop and a conditional `if` statement to find the town that got 15 inches of snowfall.

```
# Initializing a dictionary of snowfall totals for various towns
snowfall_mass_jan = {"Boston": 24.5, "Brookline": 15, "Cambridge": 14, "Framingham": 12.2,
"Malden": 20, "Wakefield": 21.2, "Norwood": 19.5}
```

```
# Finding the town that got 15 inches of snowfall
for town in snowfall_mass_jan:
    if snowfall_mass_jan[town] == 15:
        print(town)
```

Output: Brookline

This is an example of a linear search algorithm because it checks each item in the dictionary sequentially. More advanced types of searching algorithms include [binary search](#), [interpolation search](#), and [jump search](#).

Learn More

Algorithms are used in computer programming for a wide variety of purposes. See the below resources to learn more.

Digital Integration Teaching Initiative

- Gupta, Antrixsh. "[Introduction to Data Structures and Algorithms in Python.](#)" *Medium*, 4 Apr. 2023. Accessed 16 July 2024.
- "[What is supervised learning?](#)" *IBM*. Accessed 16 July 2024.
- "[What is unsupervised learning?](#)" *IBM*. Accessed 16 July 2024.
- Stahl, Alexander. "[Understanding The Basics: Generative AI.](#)" *Medium*, 23 Jan. 2024. Accessed 16 July 2024.